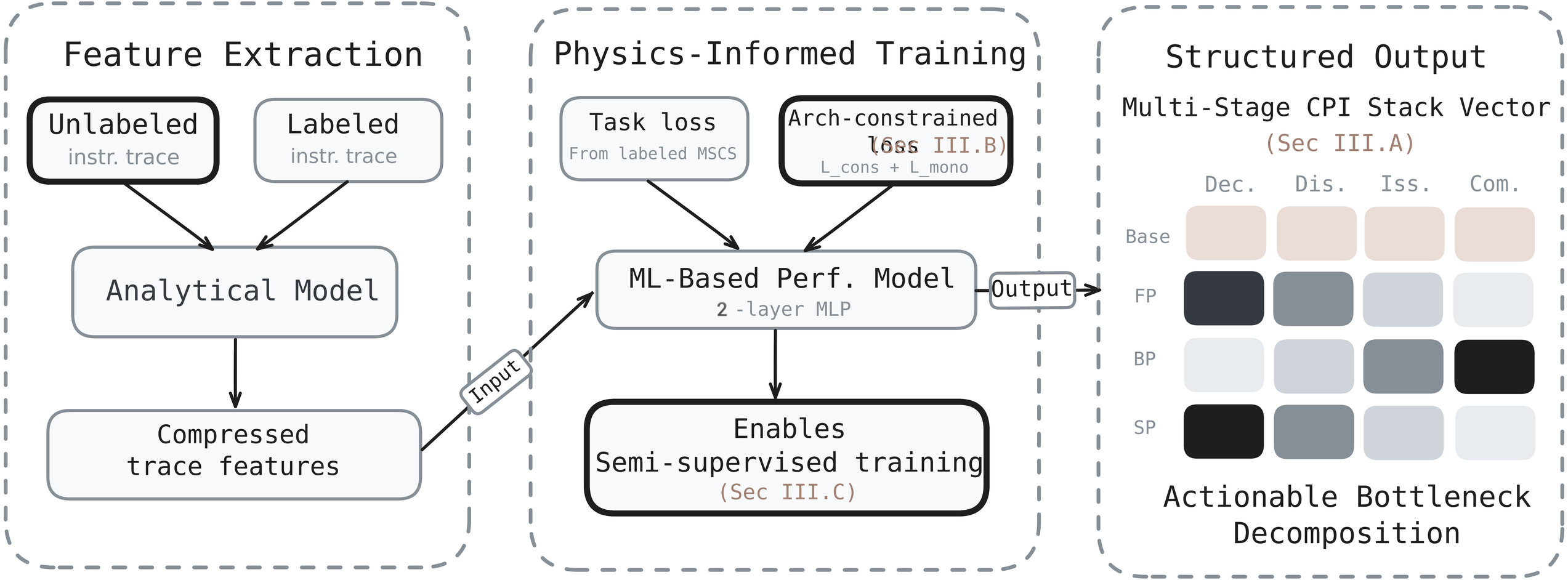
Feature Extraction
Unlabeled
instr. trace
Labeled
instr. trace
Analytical Model
Compressed
trace features
Input
Physics-Informed Training
Task loss
From labeled MSCS
Arch-constrained loss
(Sec III.B)
L_cons + L_mono
ML-Based Perf. Model
2 -layer MLP
Output
Enables
Semi-supervised training
(Sec III.C)
Structured Output
Multi-Stage CPI Stack Vector
(Sec III.A)
Dec.
Dis.
Iss.
Com.
Base
FP
BP
SP
Actionable Bottleneck
Decomposition